

## Density-Aware Modality Routing

Mengzi-BERT: density  $L_0-L_4$ , image-dep.  $R_1-R_3$   
→ budget  $N_t/N_i$ , fusion  $w$  (re-run each round)

### 1 Dual-Path Retrieval

#### Query

$q$ : text + options  
image  $I_q$

#### Text Path

BGE-M3 · top-20

#### Image Path

Qwen3-VL · top-20

#### Qwen Reranker

pool  $C$ :  $N_t + N_i$  cand.  
(prior order)

### 2 Cross-Modal Evidence Agent

vision-language policy  $\pi_\theta$  · RL-trained

judge each candidate (image + text):



keep → accept  $E$  · none useful → rewrite  $q'$

reward =  $P_G(a^* | E) - P_G(a^* | \emptyset)$

answer-utility reward (training only)

### 3 VLM Generation

image + text fed unaltered

#### Answer

A B C D

predicted  $\hat{a}$

rewrite query  $q'$  · re-retrieve